

Individualized Connectivity-Guided Versus Conventional Targeting of Accelerated Theta-Burst Stimulation in Depression: A Randomized, Double-Blind, Parallel-Design Trial

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Objective: Scalp-based repetitive transcranial magnetic stimulation targeting the left dorsolateral prefrontal cortex (DLPFC) does not account for interindividual variability. This trial was designed to determine whether individualized connectivity-guided intermittent theta-burst stimulation (iTBS) improves antidepressant outcomes compared with 5-cm targeting.

Methods: In this single-center, randomized, double-blind, three-arm trial (February 2023–March 2025), adults ages 18–65 years with major depressive disorder or bipolar II depression (≥ 1 antidepressant failure) were randomly assigned to receive 20 sessions of iTBS over 2 weeks using robotic neuronavigation with one of three targeting strategies: 5-cm rule, functional connectivity (FC)–guided targeting showing negative resting-state connectivity with the subgenual anterior cingulate cortex (sgACC), or structural connectivity (SC)–guided sites defined by probabilistic tractography to the sgACC. The primary outcome was percentage reduction in 17-item Hamilton Depression Rating Scale (HAM-D) score at week 2. Secondary outcomes included HAM-D score reduction at weeks 6 and 12, response or remission, self-reported symptoms, performance on a cognitive battery, and adverse events.

Results: Of 123 randomized participants, 119 were included in the modified intention-to-treat analyses (5-cm, N=40; SC-guided, N=39; FC-guided, N=40). SC-guided iTBS produced significantly greater HAM-D score reduction than the 5-cm rule at week 2 (least squares mean difference [LSMD]=8.79 percentage points, 95% CI=2.19, 15.40; Cohen's $d=0.70$). At week 6, both SC-guided and FC-guided groups showed significantly greater improvement than the 5-cm group (SC-guided LSMD=12.87 percentage points, 95% CI=6.21, 19.54; Cohen's $d=1.03$; and FC-guided LSMD=9.41 percentage points, 95% CI=2.84, 15.98; Cohen's $d=0.75$). By week 12, between-group differences were no longer significant. Adverse events were comparable across groups, with no seizures or mania.

Conclusions: SC-guided iTBS produced promising preliminary evidence of improved antidepressant response compared with 5-cm targeting, supporting sgACC-based connectivity-guided precision neuromodulation in depression.

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Depressive disorders are prevalent and disabling, and current treatments remain suboptimal for many patients (1). Repetitive transcranial magnetic stimulation (rTMS) is an effective noninvasive treatment for depression, including treatment-resistant cases (2), but clinical outcomes vary across individuals (3). Traditional rTMS typically relies on surface-based, one-size-fits-all targeting of the left dorsolateral prefrontal cortex (DLPFC), such as the 5-cm rule or F3 method. Although simple and widely used, these methods lack anatomical precision and fail to account for interindividual variability (4).

Individualized targeting strategies have been proposed to address these shortcomings. Anatomical imaging-guided approaches, such as targeting the boundary region between Brodmann areas 9 and 46, may improve treatment efficacy (5). More recently, circuit-informed targeting has gained attention. In particular, more negative functional connectivity (FC) between the left DLPFC and the subgenual anterior cingulate cortex (sgACC) has been associated with better antidepressant response to rTMS (6–8), prompting the hypothesis that stimulating the most anticorrelated left DLPFC site may improve outcomes. Open-label studies have produced

promising findings (9), and in randomized controlled trials, connectivity-guided rTMS has shown significantly greater antidepressant effects than sham stimulation (10, 11).

However, findings from open-label and active-sham trials leave open the question of whether such connectivity-guided approaches yield superior outcomes over standard targeting strategies. A relatively large randomized controlled trial found that left DLPFC targeting based on connectivity from the right insula did not improve outcomes over neuronavigated rTMS at the F3 target site (12). Importantly, no active-control comparison has evaluated sgACC-FC-informed targeting strategies, leaving a critical evidence gap.

Moreover, while FC-based approaches have dominated individualized rTMS, structural connectivity (SC)-based strategies also show promise (13). Diffusion tractography suggests that TMS effects propagate along white matter pathways (14), and an sgACC-connected frontal target has been shown to be feasible and to modulate neural activity in healthy individuals (13). Retrospective studies have further linked prefrontal-cingulate SC properties to antidepressant response (15). However, prospective evidence for SC-guided targeting in depression remains limited, and direct comparisons among SC-guided, FC-guided, and surface-based approaches are also lacking, and are critically needed.

To fill this gap, we conducted a randomized trial comparing FC-guided and SC-guided targeting with the 5-cm approach. We hypothesized that connectivity-guided stimulation would produce greater reductions in depressive symptoms.

METHODS

Study Design and Participants

This study was conducted at Peking University Sixth Hospital from February 2023 to March 2025. Outpatients and inpatients were screened sequentially, and eligible participants were randomly assigned to three groups. The trial was approved by the Ethics Review Committee of Peking University Sixth Hospital and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants before enrollment.

Eligible participants were 18–65 years of age; met *DSM-IV* criteria for major depressive disorder or bipolar II disorder depressive episodes, as assessed with the Mini International Neuropsychiatric Interview; had a score ≥ 18 on the 17-item Hamilton Depression Rating Scale (HAM-D); and had a history of inadequate clinical response to at least one antidepressant trial of adequate dose and duration, as determined by the treating clinician.

Key exclusion criteria included other psychiatric disorders; severe or unstable medical conditions; MRI or TMS contraindications; severe suicide risk (a score ≥ 17 on the Self-Rating Idea of Suicide Scale or suicidal behavior within 6 months); concurrent trial participation; and transcranial electrical stimulation, electroconvulsive therapy, or TMS within the past year.

Randomization

Participants were randomized in a 1:1:1 ratio to the SC-guided, FC-guided, or 5-cm group. Randomization was stratified by sex and baseline severity of depression (HAM-D score ≤ 24 versus >24) using variable block sizes and implemented through the Brainnetome-Guided Neurosurgical Robot system to ensure allocation concealment. Participants and investigators remained blinded to group assignment (see the Supplementary Methods section in the online supplement).

Targeting Procedures

All participants underwent the same robot-assisted neuronavigation and individualized MRI-guided hand motor hotspot (hMHS) localization. Briefly, the hMHS was identified using a T1-weighted MRI-guided scalp search array centered on the projection of the centroid of Brainnetome Atlas (16) region A4ul (see the Supplementary Methods section and Figure S1 in the online supplement). The 5-cm target was then calculated as the scalp point 5 cm anterior to the MRI-guided hMHS; the coordinate was recorded, and treatment was delivered under neuronavigation guidance throughout.

For SC-guided targeting, diffusion tractography was used to estimate SC between the left sgACC and left DLPFC. Diffusion-weighted images were preprocessed using MRtrix3 (17). Whole-brain probabilistic tractography was performed, followed by SIFT2 streamline weighting. Streamlines intersecting the left sgACC (A32sg in the Brainnetome Atlas; see Figure S2 in the online supplement) were selected to generate voxel-wise SC maps. Within the left DLPFC (Brainnetome Atlas regions A8dl, A9l, A9/46d, A46, A9/46v, and A8vl; see Figure S2 in the online supplement), voxels ranked in the top 5% of SC strength were retained and clustered using 26-neighborhood connectivity. The centroid of the strongest cluster was mapped to the nearest point on the scalp surface mesh as the physical SC-guided stimulation site (see the Supplementary Methods section and Figures S3–S5 in the online supplement).

For FC-guided targeting, resting-state functional MRI data were preprocessed with fMRIPrep (18), followed by nuisance regression, detrending, and band-pass filtering. The mean time series was extracted from the same sgACC seed (A32sg in the Brainnetome Atlas), and whole-brain voxel-wise FC maps were generated and Fisher z -transformed. Within the same left DLPFC mask, voxels with the most negative connectivity (lowest 5%) were retained and clustered using 26-neighborhood connectivity; the centroid of the most anticorrelated cluster was mapped to the nearest point on the scalp surface used as the FC-guided target (see Figures F2–S4 in the online supplement).

rTMS Protocol

All participants received intermittent theta-burst stimulation (iTBS) with a Magstim Rapid2 and a 70-mm figure-of-eight Alpha coil (see Figure S6 in the online supplement). An MRI-guided robotic neuronavigation system (RC-M2N2-Duet; REALControl, Ltd., China) maintained coil alignment, with stimulation delivered in a posterior-anterior orientation

(PA45°) relative to the cortical surface. The robotic arm dynamically adjusted coil position to maintain alignment, and coil-scalp contact and tracking were visually verified as needed.

Treatment consisted of 20 sessions delivered over 2 weeks, with two sessions per day spaced 2 hours apart. Each session followed an iTBS protocol (19) consisting of 1,800 pulses. Stimulation was delivered at 90% of individual resting motor threshold (RMT). The stimulation intensity was adjusted as $0.9 \times \text{RMT} + 3 \times (D_x - D_{MI})$ (20), where D_x denotes the scalp-to-cortex distance at the stimulation site, and D_{MI} the scalp-to-cortex distance at the hMHS. Participants who were initially unable to tolerate the target intensity were gradually ramped up to the full dose.

Outcomes

The primary outcome was the percentage reduction in HAM-D score at week 2, defined as $(\text{HAM-D}_{\text{baseline}} - \text{HAM-D}_{\text{week2}}) / \text{HAM-D}_{\text{baseline}} \times 100\%$.

Secondary outcomes included reduction in HAM-D score at weeks 6 and 12; rates of response ($\geq 50\%$ reduction in HAM-D score) and remission (HAM-D score < 8); scores on the Self-Rating Depression Scale; cognitive performance on the THINC-it battery (21); and neural activity changes measured by the z-transformed fractional amplitude of low-frequency fluctuation (zfALFF), a robust measure of resting-state activity.

Adverse events were assessed at each session; only clinically significant or treatment-limiting discomfort was recorded. The Young Mania Rating Scale (YMRS) was used to assess manic symptoms in participants with bipolar disorder.

Robotic system accuracy was assessed as positional and angular deviations between planned and delivered coil locations of each burst, and efficiency as the mean procedure duration for the first seven participants.

Sample Size Estimation

Based on prior studies, symptom reduction was expected to reach 62.0% in the sgACC-based FC-guided iTBS group (10) and 40.0% in the 5-cm group (22). Assuming a standard deviation of 30%, a two-tailed test with an alpha of 0.05, and 85% power, 32 participants per group (total $N=96$) were required. Assuming a 20% dropout rate, the recruitment target was 120. The FC-guided effect size estimate was used as the planning reference for both connectivity-guided arms, given their shared circuit-based rationale. The final inferential framework specified two primary pairwise comparisons (FC-guided versus 5 cm and SC-guided versus 5 cm), tested with a Bonferroni-adjusted two-sided alpha of 0.025; the FC-guided versus SC-guided comparison was exploratory.

Statistical Analysis

The primary analyses used a modified intention-to-treat (mITT) population of randomized participants who remained eligible and initiated treatment (23); four postrandomization exclusions (24) distinguished this population from the ITT population.

For the primary outcome, we used a mixed-effects model for repeated measures (MMRM) with maximum likelihood estimation, including fixed effects for group, time, and group-by-time interaction; baseline HAM-D score as a covariate; and a random intercept for participant. Secondary continuous outcomes were analyzed using the same MMRM framework. Pairwise between-group contrasts at each time point were summarized as least-squares mean difference (LSMD) with 95% confidence intervals and p values; Cohen's d was calculated using model residual standard deviation.

Sensitivity analyses included unadjusted analyses, complete analyses, extended covariate adjustment, exclusion of participants with recent medication changes, alternative outcome parameterizations, and analyses of covariance (ANCOVAs) with multiple imputation for missing end-point data.

Cognitive outcomes (available at baseline and week 2 only) were analyzed using ANCOVA with the week-2 score as the dependent variable and the baseline score as a covariate. Categorical outcomes were compared using chi-square or Fisher's exact tests.

Consistent with the prespecified primary contrast framework, Bonferroni correction was applied to control for multiplicity across the two primary comparisons (FC-guided versus 5 cm and SC-guided versus 5 cm) in primary and secondary outcomes, with statistical significance defined as a two-sided $p < 0.025$.

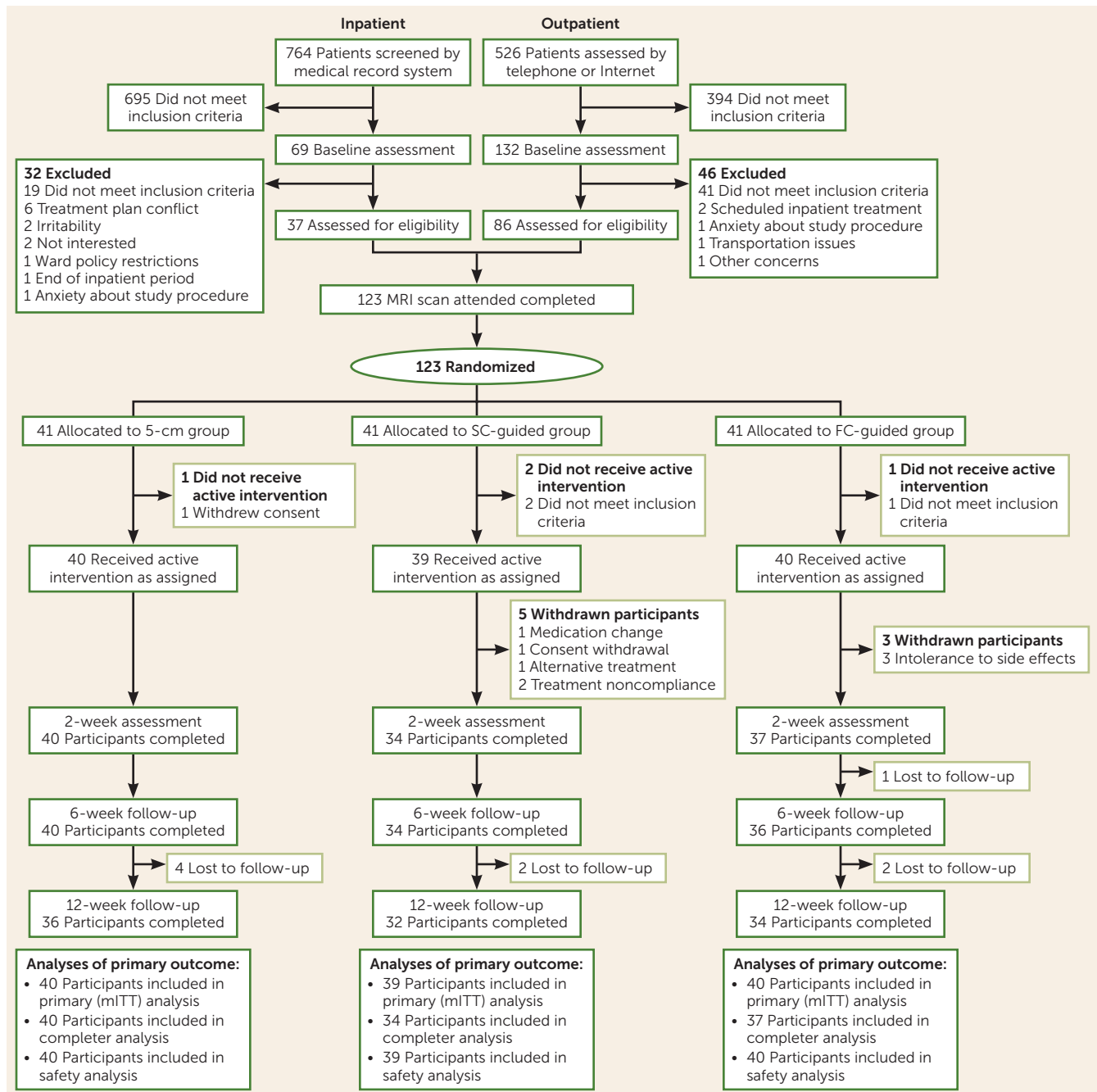
As for exploratory analyses, intraindividual Euclidean distances were calculated between the actual and computed target coordinates in individual space. Baseline FC and SC between the left sgACC and each stimulation site were computed using the same preprocessing pipeline as for target generation, with stimulation sites defined as 5-mm-radius and 10-mm-radius spheres centered on the volumetric centroid. Associations between these metrics and clinical outcomes at weeks 2, 6, and 12 were examined using Pearson, Spearman, or partial correlations; the Benjamini-Hochberg false discovery rate (FDR) correction was applied to account for multiple testing across time points, with statistical significance defined as $q < 0.05$. Analyses were conducted using SPSS Statistics, version 26.0, and R, version 4.2.1.

RESULTS

Patient Characteristics

Recruitment and participant flow are presented in Figure 1. Of 1,290 prescreened and 201 screened volunteers, 123 participants were randomized. Four were excluded before treatment because of postrandomization confirmation of ineligibility or withdrawal of consent (see Table S1 in the online supplement), leaving 40 in the 5-cm group, 39 in the SC-guided group, and 40 in the FC-guided group for mITT analysis. The mean age was 29.8 years ($SD=9.8$), the mean baseline HAM-D score was 22.2 (range, 18–36), and 21 participants (18%) had bipolar II depression.

FIGURE 1. Study flow diagram^a



^a FC, functional connectivity; mITT, modified intention to treat; SC, structural connectivity.

Baseline demographic and clinical characteristics were similar across treatment groups and generally balanced between the major depressive disorder and bipolar disorder subgroups (Table 1; see also Table S2 in the online supplement).

Concomitant medications are summarized in Table S3 in the online supplement. The mean number of failed adequate antidepressant trials during the current depressive episode was 2.14 (SD=1.50). Medication adjustments within 4 weeks before baseline occurred in 18 participants

(15%) and did not differ by treatment group. During iTBS, medication regimens were kept stable. During follow-up, treatment changes were reported by nine participants at week 6 and by four participants at week 12, with no between-group differences.

Eight participants (7%) did not complete the 2-week iTBS course (zero in the 5-cm, five in the SC-guided, and three in the FC-guided group); baseline characteristics did not differ between completers and noncompleters (see Table S4 in the online supplement).

TABLE 1. Baseline characteristics and clinical outcomes^a

Variable	5 cm (N=40)		SC guided (N=39)		FC guided (N=40)		p
	N	%	N	%	N	%	
Female	28	70	25	64	29	73	0.711
Bipolar disorder	8	20	8	21	5	13	0.576
Inpatient	12	30	11	28	13	33	0.917
Family history	14	35	14	36	12	30	0.835
Mood stabilizer use	5	13	8	21	6	15	0.610
Benzodiazepine use	7	18	8	21	10	25	0.712
Atypical antipsychotic use	14	35	10	25	15	38	0.498
Psychotherapy during rTMS	11	28	12	31	10	25	0.848
Dropped out	0	0	5	13	3	8	0.048
	Mean	SD	Mean	SD	Mean	SD	p
Age (years)	30.4	10.0	28.8	9.5	30.2	10.1	0.741
Education (years)	15.7	3.3	15.6	2.4	15.2	3.2	0.760
Age at onset (years)	23.4	9.4	23	9.4	23.1	9.1	0.985
	Median	IQR	Median	IQR	Median	IQR	p
Duration of illness (months)	76	43–128	60	36–108	72	42–108	0.566
	Mean	SD	Mean	SD	Mean	SD	p
Episodes (N)	2.2	1.1	2.2	1.5	2.2	1.6	0.980
Previous treatment history (N)	2.3	1.6	2.1	1.6	2.1	1.4	0.798
Motor threshold (% MSO)	58.4	10.8	59.6	8.5	61.5	9.2	0.478
Treatment intensity (% MSO)	48.1	8.1	44.3	7.1	48.3	8	0.010
Fluoxetine equivalent (mg)	39.1	23.4	37.9	26.1	37.2	33.3	0.952
Baseline HAM-A score	20.0	5.0	20.9	5.1	20.4	5.9	0.724
Baseline SDS score	56.2	7.7	57.6	7.6	56.8	8.2	0.725
Baseline SIOSS score	13.5	3.3	13.1	3.4	13.3	3.2	0.782
	Median	IQR	Median	IQR	Median	IQR	p
HAM-D score							
Baseline	21.5	19–24	21.0	19–24	21.0	19–26	0.991
Week 1	16.0	12–18	16.0	14–18	15.0	12–19	0.747
Week 2	12.0	8–15	8.5	6–13	11.0	7–15	0.150
Week 6	10.0	7–13	7.5	5–11	8.0	5–12	0.015
Week 12	8.0	7–13	7.0	6–11	8.0	7–12	0.950

^a FC, functional connectivity; HAM-A, Hamilton Anxiety Rating Scale; HAM-D, Hamilton Depression Rating Scale; IQR, interquartile range; MSO, maximum stimulator output; rTMS, repetitive transcranial magnetic stimulation; SC, structural connectivity; SDS, Self-Rating Depression Scale; SIOSS, Self-Rating Idea of Suicide Scale.

Outcomes

Depressive symptom trajectories were evaluated using HAM-D score reduction from baseline (Figure 2). The MMRM analysis revealed significant group-by-time interactions ($F=2.41$, $p=0.015$) and time effect ($F=289.97$, $p<0.001$) but not group effect ($F=2.06$, $p=0.132$).

Primary outcome. At week 2, SC-guided stimulation produced greater HAM-D reduction than the 5-cm rule (LSMD=8.79 percentage points, 95% CI=2.19, 15.40, $p=0.010$, surviving Bonferroni correction; Cohen's $d=0.70$), whereas FC-guided stimulation did not differ significantly from stimulation guided by the 5-cm rule ($p=0.212$) (Table 2).

Secondary outcomes. At week 6, both the SC-guided and FC-guided groups achieved greater HAM-D score reductions than the 5-cm group (SC versus 5 cm: LSMD=12.87 percentage points, 95% CI=6.21, 19.54, $p<0.001$; Cohen's $d=1.03$; FC

versus 5 cm: LSMD=9.41 percentage points, 95% CI=2.84, 15.98, $p=0.005$; Cohen's $d=0.75$), and both of these differences remained significant after Bonferroni correction. By week 12, no significant differences were observed.

SDS scores, response and remission rates, and cognitive performance did not differ significantly among groups at any time point (Table 2; see also Tables S5–S8 and Figure S7 in the online supplement).

Between the two connectivity-guided groups, no significant differences in HAM-D score reduction were observed at any time. The SC-guided group showed slightly greater improvement in self-reported depressive symptoms at week 2 (unadjusted $p=0.049$, Cohen's $d=0.70$) and faster reaction time on the Spotter task compared to the FC-guided group (unadjusted $p=0.016$, Cohen's $d=0.64$).

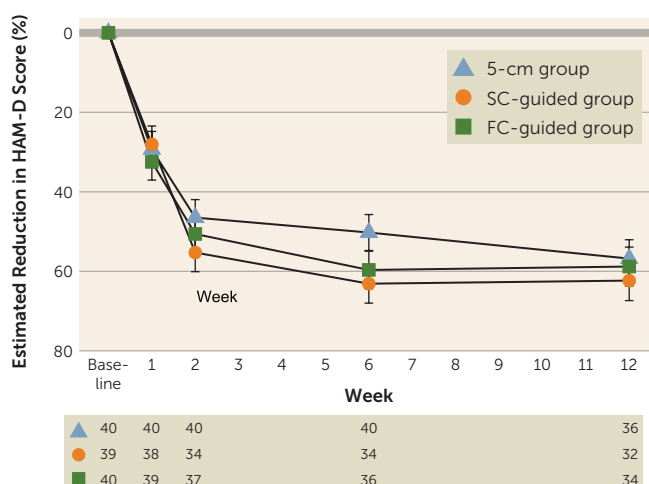
Safety

Mean stimulation intensity was 46.5% of maximum stimulator output (approximately 79% of RMT),

reflecting shorter scalp-to-cortex distances at the left DLPFC. The incidence of adverse events did not differ significantly among the three groups ($\chi^2=0.76$, $p=0.685$). Most participants reported no adverse events: 85% in the 5-cm group, 82% in the SC-guided group, and 78% in the FC-guided group. The most commonly reported adverse events were local discomfort, headache, fatigue, and dizziness. Other adverse events were rare (see Table S9 in the online supplement), and no seizures or manic episodes occurred.

Exploratory Analyses: Spatial Targeting Profiles and Neural Correlates of Response

SC-guided targets clustered anteriorly in the left DLPFC with low variability, 5-cm targets were more posterior, and FC-guided targets were more dispersed across anterior-lateral regions (Figure 3A; see also Figures S8 and S9 in the online supplement). In the FC-guided and 5-cm groups,

FIGURE 2. Percentage reduction in HAM-D scores over time^a


^a Data are presented as least-square means with 95% confidence intervals. Sample sizes for each group at each time point are shown beneath the x-axis. FC, functional connectivity; HAM-D, 17-item Hamilton Depression Rating Scale; SC, structural connectivity.

shorter distance to computed SC-guided target was associated with greater HAM-D score reductions at week 12 (median distance=42.36 mm, IQR=27.26–48.86; $r=-0.310$, FDR-corrected $q=0.027$) (Figure 3B), with a similar trend at week 6 ($r=-0.241$, $q=0.054$). In the combined SC-guided and 5-cm samples, distance to the computed FC-guided target was not associated with efficacy (median=26.50 mm, IQR=20.37–35.11). However, in the 5-cm group alone, shorter distance was associated with greater reductions at week 12 ($r=-0.421$, $q=0.032$; see Figure S10 in the online supplement).

FC-guided sites showed the most negative FC with the sgACC at baseline, while SC-guided sites showed the strongest SC (see Figure S11 in the online supplement). Across the full sample, stronger stimulation site–sgACC SC was correlated with greater reductions in HAM-D score at week 2 ($r=-0.236$, $q=0.019$) and week 6 ($r=-0.296$, $q=0.005$), whereas baseline FC was not associated with outcome (Figure 3C; see also Figures S12 and S13 in the online supplement). Within the SC-guided group, zfALFF decreases within the sgACC were associated with symptom improvement at week 2 (partial $r=0.408$, unadjusted $p=0.031$) and week 6 (partial $r=0.434$, unadjusted $p=0.021$) after covariate adjustment but did not survive FDR correction (see Figure S14 in the online supplement).

Accuracy and Efficiency of the Neuronavigation Robot

Targeting deviations were continuously recorded during each stimulation burst. The median positional deviation was 0.39 mm (IQR=0.28–0.56), and the median angular deviation was 0.17° (IQR=0.15–0.23). In the first seven participants, mean preparation time was 39.6 minutes for the initial session and 4.5 minutes for subsequent sessions (see Table S10 in the online supplement).

Sensitivity Analysis

Completer analysis results were consistent with those for the mITT sample, including the significant group-by-time interaction ($F=2.85$, $p=0.010$; see Table S8 in the online supplement). Adjusting for patient characteristics and targeting deviations did not change the primary outcome (see Table S11 in the online supplement). Findings were also consistent after exclusion of participants with prebaseline medication changes and censoring of follow-up observations after reported treatment changes (see Table S12 in the online supplement).

The primary outcome was robust across analyses based on alternative HAM-D outcome definitions (absolute change and observed scores), an alternative baseline-adjusted ANCOVA approach, and multiple imputation for missing data (see Tables S13–S15 in the online supplement).

As for diagnosis (major depressive disorder or bipolar disorder), no significant interaction with treatment group was detected ($F=0.10$, $p=0.910$; see Table S16 in the online supplement). Although diagnosis-stratified analyses were underpowered, the efficacy pattern across groups was broadly similar (see Tables S17–S19 in the online supplement).

DISCUSSION

This prospective, double-blind, three-arm randomized controlled trial directly compared two sgACC-based targeting strategies (SC-guided and FC-guided) with the 5-cm approach under identical iTBS parameters. All groups showed clinically meaningful improvement in depressive symptoms, with distinct response trajectories and no serious adverse events.

Relative to the 5-cm protocol, sgACC-based connectivity-guided stimulation showed modest advantages, with greater improvement in the SC-guided group at week 2, and in both the SC- and FC-guided groups at week 6. These between-group differences remained robust after imputation for missing data, although they were smaller than those reported in sham-controlled trials (10, 11), likely in part because the 5-cm approach is an established active comparator. In contrast, the BRIGHtMIND trial (12) reported no superiority of insula-guided iTBS over MRI-guided F3-rTMS, although differences in stimulation modalities and intensities complicate interpretation of targeting effects. More broadly, comparator design, targeted circuits, and dose-response effects may all influence observed effect sizes (25).

The accelerated response in the SC-guided group suggests that structural pathways may provide an anatomical substrate for more efficient downstream engagement of depression-relevant circuits (14). Supporting this, participants in the other groups showed greater improvement when their stimulation sites were incidentally closer to their computed SC-guided targets. Furthermore, tractography-informed targeting yielded reliable, dose-dependent engagement of the subgenual region (13), and fronto-cingulate SC has been linked to the antidepressant effect of rTMS (15). Consistently, higher baseline SC was associated with greater symptom improvement in our

TABLE 2. Primary and secondary outcomes^a

Time	5 cm			SC guided			FC guided			SC vs. 5 cm			FC vs. 5 cm			SC vs. FC		
	Mean	SE		Mean	SE		Mean	SE		Diff	95% CI	p	Diff	95% CI	p	Diff	95% CI	p
Estimated HAM-D reduction (%)																		
Week 1	29.3	2.3		28.1	2.3		32.5	2.3		-1.3	-7.7, 5.2	0.700	3.1	-3.3, 9.6	0.335	-4.4	-10.9, 2.1	0.182
Week 2	46.5	2.3		55.3	2.4		50.6	2.4		8.8	2.2, 15.4	0.010	4.1	-2.4, 10.6	0.212	4.7	-2.0, 11.4	0.169
Week 6	50.2	2.3		63.1	2.5		59.7	2.4		12.9	6.2, 19.5	<0.001	9.4	2.8, 16.0	0.005	3.5	-3.4, 10.3	0.317
Week 12	56.8	2.4		62.3	2.5		58.8	2.5		5.5	-1.4, 12.5	0.116	2.0	-4.8, 8.8	0.562	3.5	-3.5, 10.6	0.321
SDS score change																		
Week 1	6.7	1.1		6.6	1.2		3.8	1.2		-0.1	-3.3, 3.1	0.953	-2.8	-6.1, 0.4	0.082	2.8	-0.6, 6.1	0.102
Week 2	9.7	1.1		11.0	1.2		7.7	1.2		1.4	-1.9, 4.6	0.413	-2.0	-5.2, 1.2	0.225	3.3	0.1, 6.7	0.049
THINC-it cognitive battery																		
Response time in Spotter task (ms)																		
Week 2	547.6	23.7		525.4	27.0		600.5	24.3		-22.2	-82.6, 38.3	0.469	52.9	-5.8, 111.7	0.077	-75.1	-136.0, -14.1	0.016
Total correct number in Codebreaker task																		
Week 2	64.8	2.1		65.5	2.4		61.9	2.1		0.6	-4.7, 5.9	0.816	-2.9	-8.0, 2.2	0.264	3.5	-1.8, 8.9	0.193
Total correct number in Symbol Check task																		
Week 2	34.0	1.1		34.9	1.3		33.3	1.2		0.9	-2.1, 3.9	0.540	-0.7	-3.6, 2.1	0.612	1.7	-1.4, 4.7	0.282
Response time in Trails task (s)																		
Week 2	17.8	1.6		20.6	1.8		20.0	1.6		2.8	-1.4, 6.9	0.191	2.1	-1.8, 6.1	0.286	0.6	-3.5, 4.7	0.767
PDQ-5-D scores (scaled $\times 10^{-3}$)																		
Week 2	1.9	0.2		1.9	0.2		1.8	0.2		0.0	-0.4, 0.4	0.906	-0.1	-0.5, 0.3	0.619	0.1	-0.3, 0.5	0.552

^a Values are model-based adjusted means and standard errors. The p values are unadjusted, and boldface p values are statistically significant after Bonferroni correction for the two primary comparisons ($\alpha=0.05/2=0.025$). Diff, adjusted mean difference between groups; HAM-D, 17-item Hamilton Depression Rating Scale; PDQ-5-D, 5-item Perceived Deficits Questionnaire; SDS, Self-Rating Depression Scale.

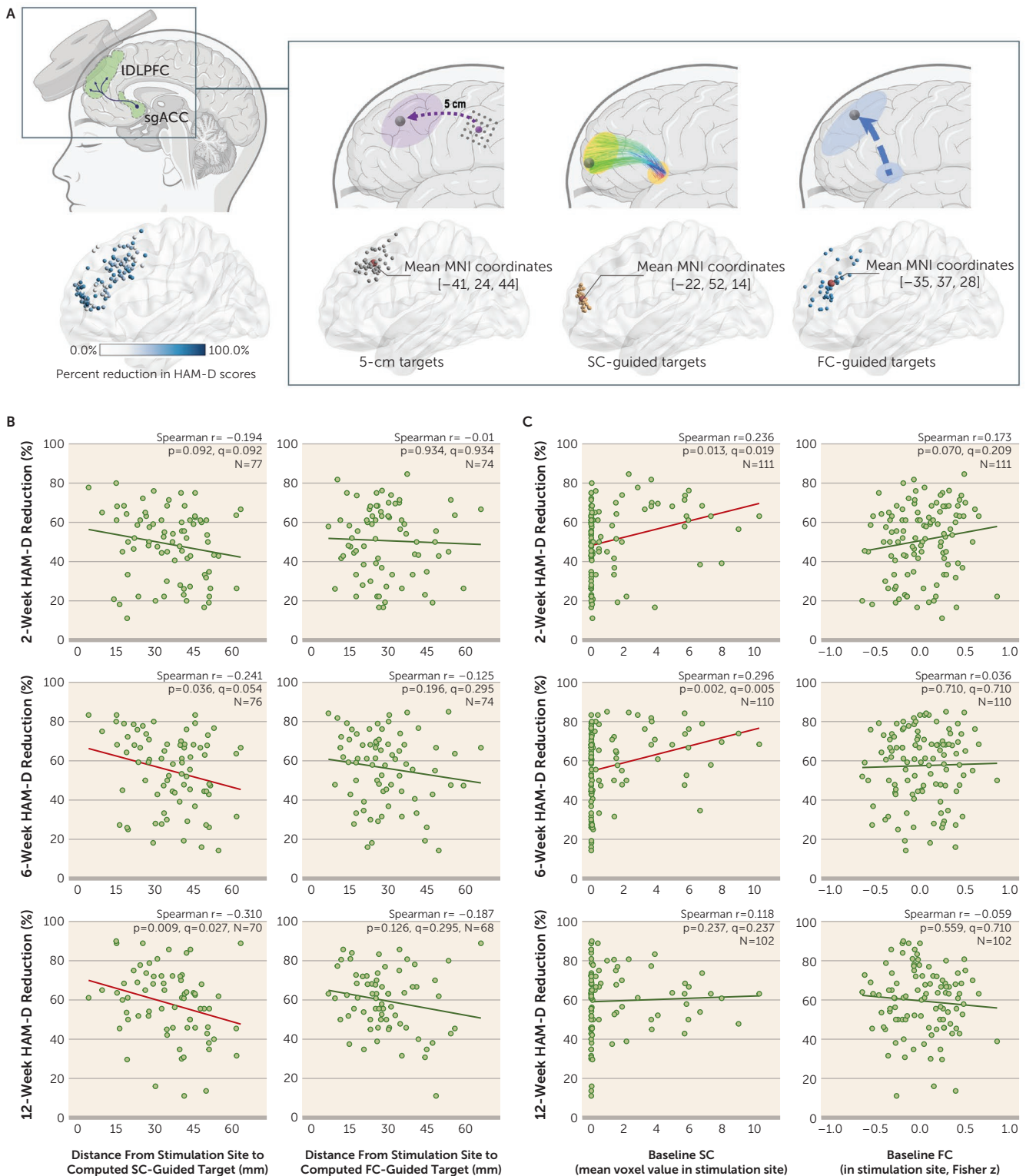
cohort, and sgACC-related functional changes following SC-guided stimulation showed nominal associations with outcomes.

A related line of work suggests that baseline left DLPFC–sgACC FC may also inform treatment response (6–8), but this signal was not robust in our cohort. This aligns with prospective routine-care evidence indicating that subgenual connectivity to the effective stimulation site has only modest predictive value (26) and is sensitive to clinical and methodological heterogeneity (27). Importantly, the limited predictive utility of baseline FC does not argue against the therapeutic potential of FC-guided targeting. The FC-guided group demonstrated greater symptom improvement than the 5-cm group by week 6. Together with prior sham-controlled trials (10, 11), these findings support the clinical promise of FC-guided approaches. Prior work has also shown that closer proximity to FC-defined targets is associated with better outcomes (28), a pattern also observed in the 5-cm group at week 12. A recent study further suggests that DLPFC sites with more negative FC to the sgACC are linked to shorter and more direct polysynaptic white matter pathways (29). Given that FC is modified by iTBS (30) and stronger functional-structural coupling within the left DLPFC may predict response (31), one interpretation is that such functional plasticity may be constrained by the underlying structural architecture.

By week 12, clinical gains were sustained across groups with no significant differences, likely reflecting delayed improvement in the 5-cm group. The 5-cm rule often engages variable functional networks across individuals and falls within a posterior transitional DLPFC region (32), which is weakly structurally connected to the sgACC. Less specific circuit engagement may contribute to the slower onset of improvement in the 5-cm group (32, 33). Furthermore, treatment was delivered as an acute course without maintenance; posttreatment management and spontaneous plasticity may have attenuated between-group differences over time. Nonetheless, the rapid improvement observed with SC- and FC-guided stimulation remains clinically meaningful, as earlier relief is associated with better long-term functional outcomes and lower relapse risk (34, 35).

Sensitivity analyses suggested that the primary findings were not significantly influenced by diagnostic heterogeneity, with no significant diagnosis-by-group interaction. Although earlier studies reported limited benefit of iTBS in bipolar depression (36), outcomes in our bipolar disorder subgroup were broadly comparable to those in the major depressive disorder subgroup, consistent with recent connectivity-guided trials (11) and meta-analytic evidence (37).

Our findings highlight the practical utility of SC-guided targeting. Relative to the higher spatial variability of FC-guided sites, SC-guided targets showed greater anatomical consistency across individuals and

FIGURE 3. Spatial distribution of stimulation targets and associations with treatment response^a


^a Panel A shows the distribution of individualized stimulation targets on the cortical surface. Colors represent the maximum percentage reduction in HAM-D score. The panel inset shows target distributions for the 5-cm, SC-guided, and FC-guided groups on the MNI template, with mean coordinates indicated. FC-guided targets showed greater spatial variability (mean Euclidean distance from group center in MNI space, 17.98 mm [SD=8.69]) than SC-guided (mean distance, 6.67 mm [SD=4.47]) and 5-cm (mean distance, 12.13 mm [SD=6.40]) targets. Panel B shows correlations between treatment response and the distance to each participant's SC-guided target (left, within 5-cm and FC groups) and FC-guided target (right, within 5-cm and SC groups). Panel C shows associations between baseline connectivity measures and treatment outcomes. Scatterplots show baseline SC and FC strength between the left DLPFC target (10-mm-radius sphere around the stimulation site) and the sgACC. Results are

clustered anteriorly within the broad left DLPFC search space, a region shown to have moderate projections to the sgACC in primate work (38). This suggests that SC-derived coordinates may provide a practical group-level reference when an advanced individual connectivity map is unavailable and may also inform refinement of scalp-based targeting approaches in settings without neuronavigation. Robot-assisted MRI-guided neuronavigation supported highly precise delivery and reduced off-target error through real-time head-motion correction, with deviations consistently below 1 mm and 1° in our trial.

Methodologically, FC-guided localization followed the protocol described by Cash et al. (39). We used a cluster-based method to define individualized targets, improving intraindividual robustness while preserving interindividual specificity. However, subject-level FC localization remains vulnerable to nonneural noise and analytic choices (27). Likewise, diffusion tensor imaging-based, SC-guided targeting is constrained by the limitations of tractography, including difficulty resolving crossing fibers and limited representation of polysynaptic communication (40). More standardized and advanced methods may further optimize target definition in future work.

Several limitations warrant acknowledgment. First, the sample size limited power for moderator analyses. Second, pharmacotherapy was not standardized; although treatments reflected real-world practice and were broadly balanced across groups, residual interaction with stimulation effects cannot be excluded. Third, inferences on longer-term durability are limited by the 12-week follow-up, and the 20-session course, while clinically standard, was relatively short and may limit our ability to detect between-group differences. Finally, although this study was designed to compare active targeting strategies, the absence of a sham control precludes fully isolating specific stimulation effects from nonspecific factors. Blinding integrity and treatment expectancy were not formally assessed, and pain was not systematically measured, limiting assessment of possible unblinding even though study procedures were intended to minimize expectancy across groups. Larger, stratified trials with optimal stimulation parameters, courses, and follow-up are needed to confirm these findings and refine personalized neuromodulation.

In summary, our findings provide promising preliminary evidence that sgACC-based connectivity-guided iTBS, particularly using structural connectivity, may accelerate antidepressant response without increasing adverse effects.

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The datasets generated during this study are available from the corresponding authors on reasonable request. Due to privacy and ethical considerations, the data are not publicly accessible.

De-identified data and analysis code are available from the corresponding authors on reasonable request under a data use agreement; raw MRI data are not publicly available, owing to privacy restrictions. Contact Dr. Yan (hao_y@bjmu.edu.cn).

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Spearman rank correlations with p values and sample sizes. Each circle indicates a participant; red regression lines denote significant associations (uncorrected $p < 0.05$). False discovery rate corrections were applied across the three time points (weeks 2, 6, and 12), and the corresponding q values are presented. DLPFC, dorsolateral prefrontal cortex; FC, functional connectivity; HAM-D, 17-item Hamilton Depression Rating Scale; MNI, Montreal Neurological Institute; SC, structural connectivity; sgACC, subgenual anterior cingulate cortex.

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